Silicosis and **asbestosis** are occupational respiratory diseases resulting from inhalation of particle of silica from sand grinding and asbestos into the respiratory tract. Workers, working in such industries must wear protective masks.

6.9 Effects of Smoking

Today due to curiosity, excitement or adventure youngsters start to smoke and later get addicted to smoking. Research says about 80% of the lung cancer is due to cigarette smoking.

Smoking is inhaling the smoke from burning tobacco. There are thousands of known chemicals which includes nicotine, tar, carbon monoxide, ammonia, sulphur–dioxide and even small quantities of arsenic. Carbon monoxide and nicotine damage the cardiovascular system and tar damages the gaseous exchange system. Nicotine is the chemical that causes addiction and is a stimulant which makes the heart beat faster and the narrowing of blood vessels results in raised blood pressure and coronary heart diseases. Presence of carbon monoxide reduces oxygen supply. Lung cancer, cancer of the mouth and larynx is more common in smokers than non–smokers. Smoking also causes cancer of the stomach, pancreas and bladder and lowers sperm count in men.
Smoking can cause lung diseases by damaging the airways and alveoli and results in emphysema and chronic bronchitis. These two diseases along with asthma are often referred as **Chronic Obstructive Pulmonary Disease** (COPD). When a person smokes, nearly 85% of the smoke released is inhaled by the smoker himself and others in the vicinity, called **passive smokers**, are also affected. Guidance or counselling should be done in such users to withdraw this habit.

Summary

The process of intake of oxygen rich air and giving out of air rich in carbon dioxide is generally called respiration. Pollutants and microorganisms are filtered from the inspired air by the hair and mucus present in the nostrils. The two main steps in the mechanism of respiration are inspiration and expiration which takes place due to pressure gradient in the atmosphere and lungs.

O_2 is transported in blood in dissolved form and is also bound to haemoglobin. One molecule of haemoglobin can bind four molecules of O_2. The Sigmoid shape of the O_2 haemoglobin dissociative curve shows increased affinity for each O_2 molecule.

CO_2 is transported in blood in dissolved form as carbamino haemoglobin and as H_2CO_3. HCO_3^- is produced in RBCs from CO_2 and water catalysed by carbonic anhydrase. Breathing is controlled by medullary respiratory centre.

Respiratory volumes and capacities indicate the amount of air inspired and expired during normal respiration. Our respiratory system can be affected by pollutants, pathogens and other chemical substances found in air. Lung cancer and emphysema cannot be cured and these diseases are common among cigarette smokers.

People at higher level than the sea level are prompted to altitude sickness as the barometric pressure is low in those regions. Surfactant, emphysema, Asthma and Dead space have been discussed. During vigorous exercise the rate of respiration increases.

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